

Examiner reports

Q1.

It was pleasing to see a variety of approaches in part (a), the most popular being to show length of AC = twice length of AD and then to use trigonometry. Many students, however, simply stated AC = twice AD with no justification. Others started from the given result and, after a circular argument, arrived back at the same result. Clearly this does not constitute proof. Students should realise that when two facts are stated (area and length) and a result is expected, both facts are likely to be needed.

The quality of explanation was poor, with great confusion between the use of upper and lower case letters. Errors such as $AB \times BC = AB^2C$, were not uncommon, indicating a lack of experience of working with geometry.

In part (b), most used $\tan A = \frac{\sin A}{\cos A}$, but cancelling of the $\sin A$ meant that only one solution could then be found.

Use of the graph function of the calculator was an acceptable method here, but a sketch and an explanation of what it showed were expected.

For part (c), the majority of students had calculated $A = 60^\circ$ at some point in this question.

Q2.

Many students were able to score at least 2 marks by rearranging the given equation and finding at least three correct solutions. Fewer students considered both the negative and positive square roots of 3 to obtain the full set of 8 correct answers. The majority of students made a significant error rearranging the given equation, $\tan^2 2\theta - 3 = 0$ becoming

$\tan^2 \theta = \frac{3}{2}$ was the most common error. Attempts made by expressing $\tan^2 2\theta$ in terms of $\sin^2 2\theta$ and $\cos^2 2\theta$ were rarely successful.

Q3.

This question was answered correctly by about half of all students. While this was a straightforward question, periodic sequences are a new topic on the specification, which may account for students' poorer performance. This question can be done very easily through the efficient use of allowed calculator functions.

The most frequently seen incorrect answers were π and 2π in roughly equal proportion; very few students selected 8.

Q4.

Part (a) was completed successfully by most students. A significant minority used

$\frac{\theta}{2\pi} \times \pi r^2$ when finding the area of the sector, possibly demonstrating a lack of practice with the standard formulae.

Part (b) tested knowledge of the Newton-Raphson method, which is new to the A-level

maths specification. Less than half of students were successful on this question, with many not realising that the method is used to solve an equation of the form $f(x) = 0$. Students who did rearrange the given equation often scored full marks.

Q10.

This question, which was the best answered question on the paper, provided most students with a very good start to the paper. In part (a), a correct expression for the perimeter was usually presented although a minority of students then lost the accuracy mark because they could not either solve the equation $r + r + 1.25r = 39$ or they did not provide a concluding statement after a verification approach using the given value for r . A large majority of students obtained the correct value for the required area of the sector in part (b). The most frequent error was to use the wrong formula, $A = r^2 \theta$, for the area of the sector.

Q11.

As expected, this question which tested the area of a triangle formula $\frac{1}{2} ab \sin C$ and the cosine rule, applied to the case of finding and using an obtuse angle, generally proved to be a challenge to all but the better students. Approximately one quarter of the students answered both parts correctly. With method marks in part (b) being available to those students who used an incorrect angle from part (a), more than half the entry scored 4 marks for Question 3, usually after using the corresponding acute angle satisfying the equation $\sin C = 5/6$. It is worth recording that a common error to find the obtuse angle in part (a) is illustrated by '90 + 56.4' rather than the correct method '180 – 56.4'.

Q12.

This question tested the basic trigonometry section of the specification. Finding the arc length and area of the sector was again generally done correctly, with students recalling and using the correct formulae. In part (c) most students were able to find an angle of 1.27 radians using the sine rule but a majority of these then failed to realise that this is an acute angle and did not subtract it from π to find the correct obtuse angle. A small minority of students chose to convert 0.8 radians to degrees before doing any calculations in part (c). They were not penalised for doing this and could score the final mark if they correctly converted the 107° back to radians.

Q13.

This was a very well answered question with a high proportion of students gaining full marks. A small number of students failed to round results to three significant figures or gave their answers in degrees. Some lost marks through sign errors when collecting the terms of the quadratic together prior to factorisation. The most fundamental error seen was where students tried to write everything in terms of $\tan x$ and occasionally the equation $8(1 + \tan x) - 2(1 + \tan^2 x) = \tan^2 x - 2$ was seen. Those who changed to sine and cosine at the start had to convert to an equation all in cosine to earn marks and some did not go far enough.

Q14.

This question provided most candidates with a very good start to the paper. The two most

common errors, made by the very small minority of candidates who failed to score full marks on the question, were to use the wrong formula, $A = r^2\theta$, for the area of the sector

or to incorrectly rearrange $21.6 = 18\theta$ to obtain $\theta = \frac{5}{6}$.

Q15.

This question, which examined trigonometry based around the cosine rule etc, was another good source of marks for most candidates. In part (a), many used the given area to convincingly show that the length of AC was 16 metres, and then, in part (b), applied the cosine rule correctly to find the correct value for BC . Some less able candidates attempted to use Pythagoras's Theorem to find BC . In part (c), it was clear that many candidates did not appreciate that the smallest angle was opposite the side with the smallest length. A significant number of solutions consisted of applying the sine rule twice to find both angle B and angle C , even in preference to using the sum of the angles of a triangle to find the size of the remaining angle.

Q16.

Most candidates had part (a)(i) correct, although the amount of work they did varied considerably. Some just wrote down the value of $\tan \alpha$, presumably recognising the 3-4-5 triangle, whereas others used Pythagoras. Several weaker candidates divided $\sin \beta$ by

$\cos \alpha$ to give an answer of $\frac{5}{6}$. Some candidates used trigonometric identities, some finding $\sin \alpha$ first and using it together with the given cosine to calculate the tangent.

In part (a)(ii), the majority of candidates failed to work with a negative value for $\tan \beta$, despite most of them showing awareness of an obtuse angle through quoting 150° . These candidates then could not get to the right answer in part (b).

However, in part (b), most candidates did use the compound angle formula for $\tan(\alpha + \beta)$ correctly, with only a few sign errors seen and some leaving the $\pm$, from the formula book, in their expression. Most tried to manipulate the surds in their expression towards the required form of the answer, but with varying degrees of success. Those who inverted the denominator and then multiplied the resulting fractions were the more successful. Some who had worked with the correct value of $\tan \beta$ failed to get to the required form by not seeing that they could cancel or multiply through their expression by a factor of $\sqrt{3}$.

Q17.

This question, which tested the basic trigonometry section of the specification, gave the value of $\sin \theta$, enabling the area of the triangle in part (a) to be evaluated simply and exactly. This was misinterpreted as a value of the angle θ by some students, for which generous allowance was made in this and the final part. Part (b) specifically asked for the exact value of $\cos \theta$, the significance of which was ignored by many. Those who used the identity $\cos^2 \theta = 1 - \sin^2 \theta$ generally obtained the correct value and full credit was also given for a correct exact value straight from the calculator. The cosine formula was generally well-known and correctly written down in part (c), but there was a substantial minority of students who used the wrong order of evaluation, which resulted in $AC^2 =$

$30.25\cos\theta$. A higher proportion of students than usual seemed to have their calculators set in the wrong mode. The final mark was reserved for those calculating the length of AC without unnecessary approximation.

Q18.

Finding the arc length and expressing the angle PTQ in radians in terms of π were generally done correctly in this question on radian measure and geometry. Most students then found the correct value for the area of the sector, but then using this proved challenging for the average grade student. A further length was needed, which was usually correctly found by more able students. Finding the area of two triangles (or the kite directly), followed by the shaded area, required connected use of information given and found; this proved to be discriminating. Those who joined the points O and T and used basic trigonometry to find the length of PT ($=TQ$) were generally more successful in finding the area of the kite than those who started by joining P to Q . There were a number of excellent solutions seen, mainly from students who supported their method by explicitly referring to unknown lengths from the given diagram.

Q19.

This question provided many candidates with a good start to the paper. Most candidates quoted the general formulae for arc length in part (a) and for area of sector in part (b), showed the correct substitution and evaluated each correctly to score full marks. There was no single common error in part (a), although wrong rearrangement of $4 = 5\theta$ to get $\theta = 1.25$ was seen and this often led to a valid follow through answer in part (b), so scoring three of the four marks for the question. Candidates who worked in degrees were generally less successful. Some weaker candidates assumed that the chord AB was also 4cm and applied the cosine rule to find the value of θ . Other than quoting incorrect formulae for the area of the sector in part (b) there were no other notable errors.

Q20.

In part (a), most candidates started by writing a correct cosine rule which they then rearranged correctly. A greater proportion of candidates than in previous series seemed to take heed of previous advice and qualified the printed answer of 97.9° with either greater accuracy or by giving its cosine value in exact form. Almost all candidates were aware of the relevant formula for the area of a triangle and a very large majority scored full marks for part (b)(i). The most

common error was to apply the formula incorrectly by writing area $= \frac{1}{2} \times 10 \times 8 \times \sin 97.9$. In part (b)(ii), those candidates who recognised the relevance of their answer to part (b)(i) produced a correct solution in a couple of lines to find the length of the perpendicular AD . However, such candidates were in a minority and many more tackled this part by using the sin e rule to find either angle B or angle C and then applying basic trigonometry to a relevant right-angled triangle. Although this was a multi-step approach there were many successful outcomes. A common wrong method was to assume that AD bisected angle BAC . Many less able candidates made no attempt to answer part (b)(ii).

Q21.

This opening question provided a confident start for most candidates with correct answers

often seen to both parts.

In part (a) most candidates recognised the relevance of the sine rule and were able to rearrange it correctly. However, giving a convincing conclusion of θ (or A) = 64° to the nearest degree was lacking in a minority of cases, either as a result of inadequate intermediate accuracy being shown, or simply the lack of an appropriate left-hand side of the equation.

In part (b) many candidates calculated the area of the triangle correctly. It was encouraging to see many of those candidates who could not answer part (a) applying good examination technique and making use of the printed result for the size of the angle in their solutions for part (b).

Q22.

There were many correct answers to parts (a) and (b)(i) with the majority of candidates having learnt the correct formulae for the area and arc length of a sector.

In part (b)(ii) most candidates found the correct values for the perimeter and k but a far higher proportion than expected of these candidates did not write a concluding statement linking the two parts and so failed to score the final mark.

Q23.

Part (a) was answered well with most candidates using the trapezium rule with the correct number of strips and values substituted correctly. The most common errors were using degrees instead of radians and incorrect use of brackets.

Most candidates recognised that the required transformation for part (b) was a stretch, but were less sure of the direction and/or scale factor.

In part (c) the most common approach was to attempt to write the given equation in terms of

$\tan x$ with varying degrees of success. Although some very good solutions were seen, a common wrong method is illustrated by ' $2\tan x = 1$, $2x = 0.785$, 3.93 , $x = 0.393$, 1.97 ,'

Those who correctly reached either $\sin^2 x = \frac{1}{5}$ or $\cos^2 x = \frac{4}{5}$ rarely scored more than one of the four marks because they forgot to consider both the positive and negative roots of these equations.

Q24.

Most candidates quoted the general formulae for the area of a sector and the arc length, showed the correct substitution and evaluated each correctly to score full marks for part (a)(i) and part (a)(ii). A significant minority of candidates, far more than anticipated, found the area of the shaded region for which no credit was awarded. Those candidates who attempted to find the perimeter almost always gained the mark for stating $PB = 5\text{cm}$.

The majority also realised that the cosine rule was required, although some incorrectly assumed that OPA was a right-angle and applied Pythagoras' Theorem to find the wrong

value for the length of AP . The usual errors in applying the cosine rule were seen: wrong order of evaluation and calculators set in the wrong mode. However, for many of the candidates, this question turned out to be their best answered on the paper.

Q25.

Part (a) was reasonably well answered, with most candidates obtaining both values. Some candidates lost the accuracy mark through inaccurate evaluation of the second

angle, with 5.94 being a common incorrect answer. A few cases of $\operatorname{cosec} x = \frac{1}{\cos x}$ were seen.

Part (b) was answered very well, with most candidates who obtained full marks in part (a) also obtaining full marks in part (b). The majority of candidates earned the first 4 marks but some then lost the final mark(s) through inaccurate values. There were a few candidates who started with the wrong identity and hence scored zero.

Q26.

In part (a), most candidates were able to find R and α correctly, with only a few arithmetic errors seen. Some candidates made the error of using $\tan \alpha = \frac{1}{3}$, or equivalent, although most candidates did use the tangent to find the value of α . Some ignored the request for three decimal places, but two decimal place accuracy was condoned.

Many candidates obtained a correct answer to part (b), although some did not give the minus sign on $\sqrt{10}$, or thought 0 or -1 was the minimum value. There was confusion among some of the candidates between the minimum value and the value of x at which it occurred, although again many correct answers were seen, but here too not always to three decimal place accuracy.

In part (c), candidates who realised that they were to use part (a) usually made some progress. Those relatively few candidates who did not realise this usually tried to square the equation, but many did not realise this could only be successful if the sin or cos term was moved to the other side first. Many such attempts were abandoned. Those who used the expected method often lost 1 or 2 marks, through not giving all possible solutions in the range, and in particular the last accuracy mark was often lost as candidates did not make use of -0.886 in their working. Three decimal place accuracy was required for the final mark as were answers in radians: degrees were condoned during the working and in part (b) but not in part (a).

Q27.

This opening question provided a confident start for most candidates, with correct answers often seen to parts (a) and (b)(i). However, part (b)(ii) was answered relatively poorly with a surprisingly large minority simply not knowing the correct formula for the circumference of a circle (or an arc-length with angle 2π). The most common wrong answer for those who recalled the correct formula for the circumference was ' $x = 42.7$ ',

obtained by poor use of the calculator in evaluating $\frac{22.7}{2\pi}$ as $\frac{22.7}{2} \times \pi$. Even some of the

best candidates lost marks in this opening question while getting high marks elsewhere.

Q28.

Full marks were awarded to more candidates in this question than any other question on the paper. Part (a) was well answered, with a pleasingly large proportion of candidates remembering to write down sufficiently accurate values to convince the examiners that evaluations of the rearranged sine rule had been carried out to justify a final statement matching the printed result requested in the question. Part (b) was less well answered with a non-included angle, either 11.5° or 115° being used instead of 18.5° with lengths 6

and 15 in the area formula $\frac{1}{2} ac \sin B$.

Q29.

This question, which tested trigonometric-graphs, identities and equations, was better answered than similar questions in some recent series. In part (a) most of the graphs presented had the correct shape with intercepts given, although the intercept $y=1$ was occasionally omitted and those on the x -axis only expressed in degrees. Part (b)(i) usually had enough to justify the given result which was then generally used in part (b)(ii). The most common error in the final part was to omit the second answer for x . It was not uncommon to see weaker candidates writing 'identities' such as ' $\cos 2x \times \cos 2x = \cos^2 4x$ '.

Q30.

The vast majority of candidates were able to quote and use the correct formulae for the area of the sector and for the arc length, although in part (b)(i) some candidates forgot to add twice the radius when finding the perimeter. Candidates who converted 0.8 radians to degrees and then used appropriate formulae for area of sector and arc length sometimes lost marks through premature approximation. Although part (b)(ii) was generally very well answered, there was a small minority of candidates who used ' 2π ' for the perimeter of the square.

Q31.

The vast majority of candidates were able to recall and use the correct formula for the area of the triangle. In part (b), candidates almost always showed correct substitutions into the cosine rule, which appears in the formulae booklet, and the majority went on to score all three marks. However, there were other candidates who, having substituted values into the cosine rule, just wrote down the printed answer '6.86'. Such candidates were only awarded one of the three marks because they failed to show sufficient working to the necessary degree of accuracy to justify the printed value for the length of BC .

In part (c), most candidates used the sine rule as their method rather than the alternative approach of using the area of triangle ABC . Incorrect rearrangement of the sine rule was seen in a minority of cases, but most other candidates gave a correct value for $\sin B$ although many then went on to find the value for B as their final answer; this further work was not penalised, but clearly time had been wasted. There were a small number of candidates who clearly had their calculators set in the wrong mode (normally radians). Such candidates did not always make good use of the printed answer in part (b) to recover, as they continued to use radians in part (c).

Q32.

Many candidates found the inverse sine of 0.8 correctly but a significant minority did not give the second solution. Parts (b)(i) and (b)(ii) were usually well answered but wrong signs, use of degrees and reversed coordinates were all seen in part (b)(ii). Part (b)(iii) was frequently not attempted or marks were lost due to an apparently careless omission of brackets. Sketches for $\sin 2x$ resulted in a spread of marks awarded. Those sketches gaining no credit usually resulted from stretching the given graph of $\sin x$ with a scale factor 2 parallel to the y -axis.

One mark was the usual award for those stretching the graph of $\sin x$ parallel to the x -axis but with scale factor 2. Those who correctly stretched $\sin x$ parallel to the x -axis with scale

factor $\frac{1}{2}$ scored at least two marks, and more often at least four marks, with a significant number failing to get the fifth mark because the x -coordinates of the maximum points were not shown.

Q33.

Some candidates clearly had little knowledge of this part of the specification, whilst others gave clear and concise solutions. Part (a) was more successful than part (b), although many made a sign error in comparing their expansion with the given expression and found

$$\tan \alpha = -3 \text{ or } \tan \alpha = \frac{1}{3}.$$

Very few errors were seen in finding the value of R , although many did not realise that $-R$

was the minimum value. This was frequently given as 0, -1 or $\frac{\pm 1}{\sqrt{10}}$, and it was often confused with finding the value of x where the minimum occurred.

Some candidates made the mistake of using $-\frac{\pi}{2}$ instead of $\frac{3\pi}{2}$ and gave a result which was out of range. Most candidates did work in radians to the requested accuracy, and some drew sketch graphs to aid their solution.

Q34.

Most candidates recognised that the cosine rule was required in part (a) but some misquoted it despite the formula being available in the formulae booklet. Those candidates who substituted values into the formula before rearranging were generally more successful than those who rearranged before substituting the given lengths. Some candidates failed to score all three marks because they did not show values to a sufficient degree of accuracy to justify the printed statement. There were some candidates who tried to verify the result by using 38.2 with $a = 8$ and $c = 7$ in a cosine rule to show that b was approximately 5. This approach could score no more than 1 mark.

Although the correct answer for the area of the triangle was seen many times in part (b), there was a significant minority of candidates who did not apply a correct formula. The

most common error was the omission of the $\frac{1}{2}$ in $\frac{1}{2} ac \sin B$.

Q35.

This question which tested the topics of length of an arc and area of a sector, but in an unstructured manner, was well received by the candidates and was indeed the best answered question on the paper. Many candidates presented a completely correct solution. The most common loss of marks was due to premature approximation in finding the value of r before substituting into a correct formula for the perimeter of the sector or

for an incorrect rearrangement of ' $33.75 = \frac{1}{2} \times 1.2r^2$ '. It was a rarity to see candidates using wrong formulae for area of sector and arc length.



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.